

## From hand-rolled boosting to the libraries that win

In [19] you built gradient boosting yourself: fit each tree to the *negative gradient* of the loss, shrink, repeat. The algorithm is finished.

What is left is **engineering** - the libraries that dominate tabular ML:

- ▶ **XGBoost** - a regularized objective + a fast, careful implementation;
- ▶ **LightGBM** - leaf-wise growth, smart sampling, blazing speed;
- ▶ **CatBoost** - categorical features done right.

Same gradient descent in function space - made **fast**, **regularized**, and **production-ready**.

# Outline

XGBoost: the regularized objective and the structure score

Engineering: why it is fast

The modern trio: LightGBM and CatBoost

Using them well

Stacking: combining different models

## XGBoost

A regularized objective, and a closed-form score for how good a tree is.

# XGBoost: extreme gradient boosting

A scalable tree-boosting system (Chen & Guestrin, 2016):

- ▶ a **regularized objective** - penalties baked into the loss;
- ▶ **approximate / histogram** split finding for large data;
- ▶ row + column **subsampling**; **sparsity-aware** splits;
- ▶ parallel split search and **GPU** support.

Often the top performer on tabular benchmarks - **if properly tuned**. Many knobs, and the defaults are rarely optimal.

## The plan, in plain words

Before any algebra, here is the whole idea:

1. **Score** how good a tree is by how much it would cut the loss - call it the *structure score*.
2. To get that score, **approximate the loss as a parabola** at the current prediction (a Newton step: use the slope *and* the curvature).
3. **Solve** that parabola for the best value in each leaf.
4. **Grow greedily**: keep a split only if it raises the score by more than the cost of adding a leaf.

The next slides just make each step precise. Hold on to this picture: **score a tree, then split to improve the score**.

## A regularized objective

At step  $m$  we add one tree  $b$  with leaf values  $c_1, \dots, c_T$ . XGBoost minimizes the loss *plus* complexity penalties:

$$\text{obj} = \sum_{i=1}^n L(y_i, f^{[m-1]}(x_i) + b(x_i)) + \underbrace{\gamma T}_{\# \text{ leaves}} + \underbrace{\frac{1}{2}\lambda \|c\|_2^2}_{L_2} + \underbrace{\alpha \|c\|_1}_{L_1}.$$

- ▶  $\gamma$  penalizes the **number of leaves** (tree size);
- ▶  $\lambda$  ( $L_2$ ) and  $\alpha$  ( $L_1$ ) shrink the **leaf values**.

Contrast [19]: plain gradient boosting put *no* penalty inside the objective. Here regularization is part of what every tree optimizes. (Software: `gamma`, `lambda`, `alpha`.)

## Second-order Taylor of the loss

Expand the loss around the current prediction  $f^{[m-1]}$ , to **second order**:

$$L(y, f^{[m-1]} + b) \approx L(y, f^{[m-1]}) + g b + \frac{1}{2} h b^2,$$

with gradient and Hessian

$$g = \frac{\partial L}{\partial f^{[m-1]}}, \quad h = \frac{\partial^2 L}{\partial (f^{[m-1]})^2}.$$

$g$  is exactly the **negative pseudo-residual** from [19]. XGBoost's one real addition is the **curvature**  $h$ . *Intuition*: picture a parabola hugging the loss - slope  $g$  points downhill, bend  $h$  says how big a step to trust (a Newton step).

## Group the objective by leaf

Drop the constant  $L(y, f^{[m-1]})$  and use that every point in leaf  $t$  shares one value  $c_t$ :

$$\text{obj} \approx \sum_{t=1}^T \left[ G_t c_t + \frac{1}{2} (H_t + \lambda) c_t^2 \right] + \gamma T,$$

$$G_t = \sum_{i \in \text{leaf } t} g_i, \quad H_t = \sum_{i \in \text{leaf } t} h_i.$$

*Intuition:* every point in a leaf gets the **same** correction  $c_t$ , so a leaf is just **one number to choose** - and the objective is now a simple quadratic in it.



## Optimal leaf value = the “structure score”

Minimizing each quadratic  $G_t c_t + \frac{1}{2}(H_t + \lambda)c_t^2$  gives the best leaf value

$$c_t^* = -\frac{G_t}{H_t + \lambda}.$$

Plugging back, the best objective for a *fixed tree shape* is the **structure score**

$$\text{obj}^* = -\frac{1}{2} \sum_{t=1}^T \frac{G_t^2}{H_t + \lambda} + \gamma T.$$

*Intuition:* a leaf's best step = total error to fix ( $G_t$ ) per unit of confidence ( $H_t + \lambda$ ); the score adds up each leaf's payoff - **lower is a better tree**.

**Numbers:**  $G_t = -3$ ,  $H_t = 4$ ,  $\lambda = 1 \Rightarrow c_t^* = -(-3)/(4+1) = 0.6$ .

## Worked bridge: the leaf value is [19]'s residual mean, braked

In [19] the round-1 **left leaf** just used the residual mean,  $-17.1$ . XGBoost's  $c^* = -G_t/(H_t + \lambda)$  turns out to give the *same* leaf, gently shrunk by  $\lambda$ .

For **squared loss** the Hessian is  $h_i = 1$ , so a leaf with  $n$  rows has

$$G_t = \sum_{\text{leaf}} -(y_i - F) = -\sum r_i, \quad H_t = n \Rightarrow c_t^* = \frac{\sum r_i}{n + \lambda} = \underbrace{\bar{r}}_{\text{[19]'s leaf}} \cdot \frac{n}{n + \lambda}.$$

Left leaf:  $n = 41$ ,  $\bar{r} = -17.1$ , so  $c^* = -17.1 \times \frac{41}{42} = -\mathbf{16.7}$  at  $\lambda = 1$  (barely braked), or  $-13.8$  at  $\lambda = 10$ . **The structure score is [19]'s by-hand step plus a built-in  $\lambda$  brake.**

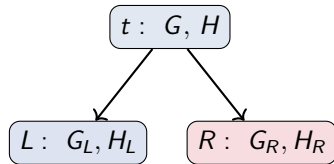
## The split-gain criterion

How good is a split? By how much it **improves the structure score**:

$$\text{gain} = \frac{1}{2} \left[ \frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right] - \gamma.$$

- *Intuition*: do the two children together score better than the parent, by more than the cost  $\gamma$  of a new leaf?
- Take the split only if  $\text{gain} > 0$ ;  $\gamma$  is the **minimum gain to keep it**.
- This - not Gini/variance - is what XGBoost maximizes at each node.

**Numbers:** parent  $(G=-3, H=4)$ ,  $\lambda=1$ , split into  $L(-3, 2)$ ,  $R(0, 2)$ :  $\text{gain} = \frac{1}{2} \left[ \frac{9}{3} + \frac{0}{3} - \frac{9}{5} \right] - \gamma = 0.6 - \gamma$  (keep only if  $\gamma < 0.6$ ); then  $c_L^*=1.0$ ,  $c_R^*=0$ .



$$G = G_L + G_R, \quad H = H_L + H_R$$

## Why it is fast

The same gradient boosting, engineered for speed and scale.

## Tree growing and pruning

- ▶ Grow each tree **level by level**, up to `max_depth`.
- ▶ A split is kept only if its **gain** (previous slide) is positive.
- ▶ After growing, **prune** back any split with  $\text{gain} < 0$  - the  $\gamma$  threshold.

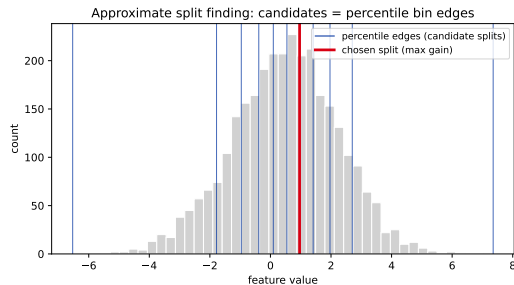
Depth-limited growth + post-pruning by gain keeps trees shallow and regularized.

# Approximate (histogram) split finding

Testing every possible split is slow. Instead, bucket each feature into  $\sim \ell$  **percentile bins** and evaluate only the **bin edges**.

- **Global**: propose the bins once per tree.
- **Local**: re-propose at every node (more accurate, more work).

“Histogram-based gradient boosting” - XGBoost `tree_method="hist"`; the default in LightGBM and sklearn `HistGradientBoosting`.



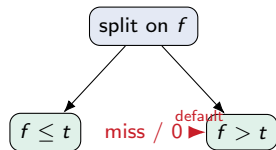
Candidates sit at percentile edges (denser where data is dense). Source: Chen & Guestrin (2016).

## Sparsity-aware splits

Real data is full of **missing values** and **zeros** (one-hot columns).

XGBoost gives every split a learned **default direction**: rows missing (or zero) on the split feature all go that way.

- ▶ Handles missing values **natively** - no imputation.
- ▶ Skips absent entries during split search, so sparse data is **faster**.



A learned default branch for absent values.

## Subsampling, parallelism, GPU

- ▶ **Row subsampling** (`subsample`) - stochastic gradient boosting.
- ▶ **Column subsampling** (`colsample_bytree/bylevel/bynode`) - like a random forest's `mtry`; faster and more diverse trees.
- ▶ Boosting is **sequential**, but the **split search inside one tree** parallelizes over sorted feature blocks - and moves well to **GPU**.

Subsampling is both a speed-up and a regularizer.



## DART: dropout for trees

Borrow **dropout** from deep learning. When adding a new tree, compute the update against a **random subset** of the existing trees (drop the rest), then rescale so the ensemble does not overshoot.



- ▶  $p_{\text{drop}} = 0$ : ordinary gradient boosting.
- ▶  $p_{\text{drop}} = 1$ : trees trained independently, equally weighted  $\approx$  a random forest.

$p_{\text{drop}}$  is a smooth dial from **boosting** to **random forest** (XGBoost booster="dart").

## The modern trio

XGBoost is the baseline. LightGBM (the favorite) and CatBoost push it further.

## LightGBM: my favorite

For everyday tabular work, this is what I reach for first:

- ▶ **very fast** and **low memory** (histogram + leaf-wise);
- ▶ excellent **out-of-the-box defaults**;
- ▶ **native categorical** support;
- ▶ scales smoothly to **large** data.

### Instructor's pick

My default first model on tabular data - the next frames are why.

Ke et al. (2017), *LightGBM*.

## Me and LightGBM ❤️



<https://youtu.be/UqmqC10ZRwg>

## Predict-first: leaf-wise vs level-wise

Give both the **same budget** of leaves. Level-wise grows a full depth level at a time; leaf-wise always splits the *highest-loss* leaf. Which **overfits sooner**?

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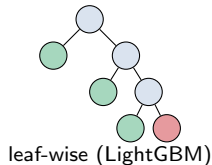
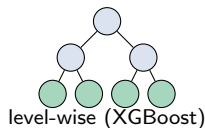
**Leaf-wise.** Spending the whole leaf budget on the worst region makes *deeper, narrower* trees that can chase noise – which is exactly why LightGBM needs `num_leaves` (and `max_depth`) kept in check.

## Leaf-wise (best-first) growth

XGBoost grows **level-wise by default** (a whole depth level at a time). LightGBM instead splits the **single leaf with the largest gain**, anywhere in the tree.

- ▶ Unbalanced, deeper trees; **converges in fewer rounds**.
- ▶ Can **overfit** - so the main complexity knob is `num_leaves` (keep it  $< 2^{\text{max\_depth}}$ ), plus `min_data_in_leaf`.

*Intuition:* spend each split where it helps most, instead of splitting every leaf just because it is at the same depth.



Source: LightGBM docs.

## Predict-first: GOSS throws away rows

GOSS keeps all **large-gradient** rows but **randomly drops most small-gradient** ones (the points the model already fits well). How much accuracy do you lose?



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**Almost none.** Well-fit rows carry little new signal, so dropping most of them (with a reweight to stay unbiased) speeds training a lot at negligible cost – that is the whole point.

## GOSS: gradient-based one-side sampling

Random subsampling wastes the **informative** rows. GOSS samples smarter, by gradient:

- ▶ keep the top  $a \cdot n$  rows with the **largest** **|gradient|** (still-hard cases);
- ▶ randomly sample  $b \cdot n$  of the rest;
- ▶ **up-weight** those sampled rows by  $\frac{1-a}{b}$  so the gain estimate stays unbiased.

*Intuition:* rows the model already gets right (tiny gradient) teach it little - spend the budget on the hard rows, but reweight the easy sample so the data still looks unbiased. Defaults  $a = 0.2$ ,  $b = 0.1$ : each split sees  $\sim 30\%$  of the rows.

Ke et al. (2017).

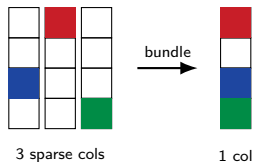
## EFB: exclusive feature bundling

In sparse, high-dimensional data many features are **mutually exclusive** - never nonzero at the same time (one-hot columns).

**Bundle** them into one feature with a **single histogram** - nearly lossless.

- ▶ Histogram cost  $\mathcal{O}(np) \rightarrow \mathcal{O}(n \cdot \text{\#bundles})$ .
- ▶ Optimal bundling is NP-hard; a **greedy** grouping works well.

*Intuition:* one-hot columns rarely fire together - pack them into one and lose almost nothing while doing far less work.



Ke et al. (2017).

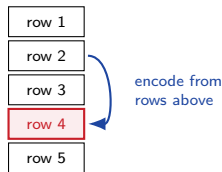
# CatBoost: the categorical specialist

Built around **categorical** features:

- ▶ **Ordered target statistics** - encode a category by its target mean on *past* rows only, so **no target leakage**;
- ▶ **Ordered boosting** - a permutation trick that removes the **prediction-shift** bias;
- ▶ **Symmetric (oblivious) trees** - one split per level: fast, regularized.

*Intuition:* the whole-dataset average peeks at the answer (leakage); a running average over *earlier* rows is honest - that is what “ordered” means.

Prokhorenkova et al. (2018).



## Which one, when?

|                       | XGBoost        | LightGBM            | CatBoost          |
|-----------------------|----------------|---------------------|-------------------|
| Tree growth           | level-wise     | leaf-wise           | symmetric         |
| Row sampling (option) | random         | random or GOSS      | random            |
| Native categoricals   | no*            | yes                 | yes (best)        |
| Speed on large $n$    | good           | <b>fastest</b>      | good              |
| Reach for it when     | solid baseline | default for tabular | many categoricals |

\* XGBoost can also grow leaf-wise and gradient-subsample, and has experimental categorical support. The table shows *defaults* - except row sampling, which is *off* by default in all three (cells name the available strategy). **All three are the same gradient boosting** - the differences are engineering and categorical handling.

## Using them well

Tuning order, domain constraints, and reading the model.

# Hyperparameters that matter

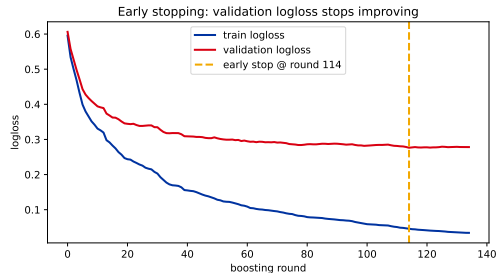
| Group      | XGBoost                          | LightGBM                              |
|------------|----------------------------------|---------------------------------------|
| Learning   | eta, n_estimators (+ early stop) | learning_rate, n_estimators           |
| Complexity | max_depth, min_child_weight      | num_leaves, min_data_in_leaf          |
| Regularize | gamma, lambda, alpha             | min_gain_to_split, lambda_l1/l2       |
| Randomness | subsample, colsample_by*         | bagging_fraction,<br>feature_fraction |

**The knob differs:** XGBoost sets tree size with max\_depth; LightGBM with num\_leaves (default 31). Defaults: XGB eta 0.3 / depth 6; LGBM lr 0.1.

# The tuning order that matters

1. Set `n_estimators` *large* and use **early stopping**.
2. Trade **learning rate** against the number of trees.
3. **Tree complexity**: `max_depth` (XGB) / `num_leaves` (LGBM).
4. **Regularization**: `gamma`, `lambda`, `alpha`.
5. **Subsampling**: rows + columns.

Search with random or Bayesian optimization (see the hyperparameter-tuning deck).



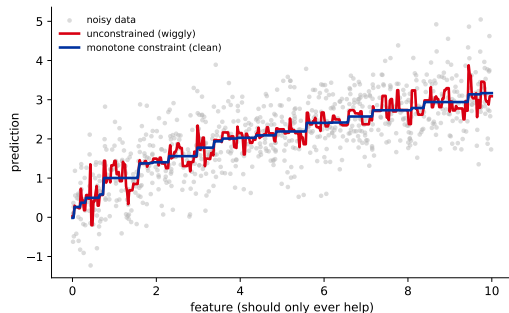
Early stopping takes the round before validation stalls  
- step 1.



# Inject domain knowledge: constraints

Sometimes you *know* something the data should obey:

- ▶ **Monotonic constraints**  
(`monotone_constraints`): force a feature's effect to be non-decreasing (or non-increasing) - e.g. "higher dose  $\Rightarrow$  no lower response".
- ▶ **Interaction constraints**: restrict which features may appear together.



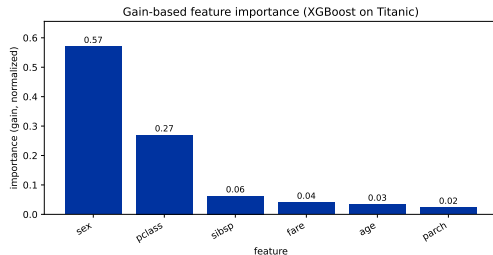
Constraints add **trust** and act as **free regularization** - use them when the direction or structure is known a priori.

# Are they black boxes? Feature importance

Boosted models are **not** opaque:

- ▶ built-in importances: **gain** (loss reduction), cover, frequency;
- ▶ but **split-count importance is biased** toward high-cardinality features - prefer **permutation importance**.

For per-prediction attribution there is **SHAP** (fast *TreeSHAP* for trees) - we will cover it in its own session.



## In code: LightGBM and XGBoost, side by side

### LightGBM (the favorite)

```
from lightgbm import LGBMClassifier
from lightgbm import early_stopping
m = LGBMClassifier(
    n_estimators=2000,
    learning_rate=0.05,
    num_leaves=31)
m.fit(X_tr, y_tr,
      eval_set=[(X_val, y_val)],
      callbacks=[early_stopping(50)])
```

### XGBoost (near-identical API)

```
from xgboost import XGBClassifier
m = XGBClassifier(
    n_estimators=2000,
    learning_rate=0.05,
    max_depth=6,
    early_stopping_rounds=50)
m.fit(X_tr, y_tr,
      eval_set=[(X_val, y_val)])
```

No install? `sklearn.ensemble.HistGradientBoostingClassifier` mirrors both.

## Beyond binary classification

The same  $g, h$  machinery powers many tasks - just change the **objective**:

- **Regression** (`reg:squarederror`), **multiclass** (`softmax`), **ranking** (`rank:*` / `LambdaMART`), Poisson, quantile - even **custom** losses (supply your own  $g$  and  $h$ ).

**Imbalanced classes?** Reweight the rare class with `scale_pos_weight` (XGBoost) or `is_unbalance / class_weight` (LightGBM), and judge with **AUC / PR**, not accuracy. Full playbook: the imbalanced-learning lecture ([15]).

## When boosting struggles

Powerful, but not magic:

- ▶ **No extrapolation** - a tree predicts a constant outside the training range; it cannot trend beyond what it has seen.
- ▶ **Needs tuning** - strong defaults exist, but top results need real HP search (a random forest is more forgiving).
- ▶ **Sequential** - trees are built one after another; less parallel than bagging.
- ▶ **Not for raw unstructured data** - images, audio, long text → neural nets; very sparse high-dimensional signals can favor linear models.
- ▶ **Miscalibrated scores** - boosted `predict_proba` is often over-confident; calibrate before you trust it (calibration lecture, [14]).

*Rule of thumb:* reach for boosting on **tabular** data; reach elsewhere when the data is not tabular.

## Stacking

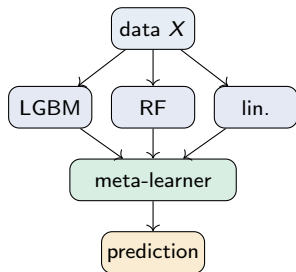
The third ensemble idea: let a meta-learner blend different model types - and close the chapter.

## Stacking: a meta-learner over base models

Bagging and boosting combine *one* kind of model. **Stacking** combines **different** ones:

- ▶ train several diverse **base learners** (level 0) - e.g. LightGBM, a random forest, a linear model;
- ▶ feed their predictions to a **meta-learner** (level 1) that learns how best to *combine* them.

*Intuition:* different models make different mistakes; the meta-learner learns whom to trust, and when.



## Stacking done right: out-of-fold predictions

**The trap:** if the meta-learner trains on base predictions of the *same* rows the base models were fit on, it sees **leaked**, over-optimistic inputs.

**The fix:** feed it **out-of-fold** (cross-validated) predictions - each row's base prediction comes from a model that did *not* train on it. StackingClassifier does this for you via cv.

```
from sklearn.ensemble import StackingClassifier
from sklearn.linear_model import LogisticRegression

base = [("lgbm", LGBMClassifier()), ("rf", RandomForestClassifier())]
clf = StackingClassifier(estimators=base,
                        final_estimator=LogisticRegression(), cv=5)
clf.fit(X_tr, y_tr)    # cv=5 -> honest out-of-fold base predictions
```



## Simpler cousins: voting and blending

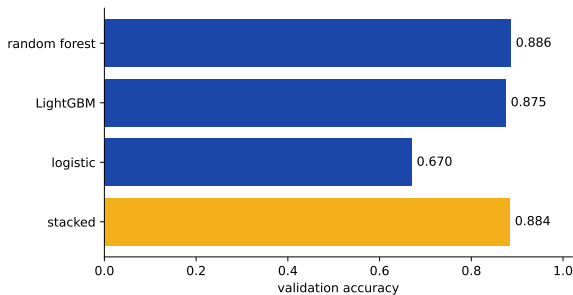
- ▶ **Voting** - no meta-learner, just combine base predictions directly: **hard** (majority class) or **soft** (average predicted probabilities, usually better).  
`VotingClassifier`.
- ▶ **Blending** - like stacking, but the meta-learner trains on a single **holdout** set instead of full cross-validation. Simpler and faster; uses a bit less data.

Stacking usually wins but costs the most (every base model + the meta-learner, with CV). Start with one strong model; stack only to chase the last bit of accuracy.

## Does stacking actually pay off?

A quick receipt on one dataset: three diverse bases, then a logistic meta-learner over their out-of-fold predictions.

Here the stack **ties** the best base (random forest) and does not beat it – the weak logistic base added no usable diversity.



The honest lesson: stacking's gains are **real but small and not guaranteed**. Start with one strong model; reach for a stack only when the bases are **diverse and each genuinely good**.

## The ensemble landscape

|             | <b>Bagging</b><br>(random forest) | <b>Boosting</b><br>(GB / XGB / LGBM) | <b>Stacking</b>           |
|-------------|-----------------------------------|--------------------------------------|---------------------------|
| Base models | same type, parallel               | same type, sequential                | different types           |
| Combine by  | averaging                         | weighted additive sum                | a meta-learner            |
| Mainly cuts | variance                          | bias                                 | both                      |
| Trains      | in parallel                       | sequentially                         | bases parallel, then meta |
| Use when    | low-tuning baseline               | top tabular accuracy                 | squeeze diverse models    |

Three ways to turn many so-so models into one strong one.

## Recap: the trees and ensembles chapter

- ▶ **One tree** ([17]) - interpretable, but unstable.
- ▶ **Bagging** → **random forests** ([18]) - average many parallel trees, cut variance.
- ▶ **Boosting** ([19]) - add trees sequentially to fix errors, cut bias; gradient descent in function space.
- ▶ **The libraries** ([20]) - XGBoost (regularized objective + structure score), **LightGBM** ❤️ (leaf-wise + GOSS + EFB), CatBoost (categoricals); tune in order, add constraints, read importance.
- ▶ **Stacking** - a meta-learner over diverse models, to squeeze out the last gains.

**Trees + ensembles dominate tabular data.** Next: models for data where trees struggle - images, text, sequences.